

Sliding Window Mode

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Your task is to calculate the mode each window of k elements, from left to right.

The mode is the most frequent element in an array. If there are several possible modes, choose the smallest of them.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print $n - k + 1$ values: the modes.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
8 3 1 2 3 2 5 2 4 4	1 2 2 2 2 4